

Technology Stack

Date	03-07-2026		
Team ID			
Project Name	Smart Agricultural Production Optimization Engine		
Maximum Marks	2 Marks		
S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

1	Front end / Client - Side	Flutter + Dart SMS Interface for feature phones	Flutter: Single codebase for Android + iOS, low cost for Govt. Offline-first: Works in rural areas with poor internet. SMS Interface: Covers farmers without smart phones using basic keypad phones.
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Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

2	Backend / Server - Side	Python + Flask/ Fast API	Python: Best for ML model integration with Opti Crop Engine. Fast API: High speed, handles 50,000 concurrent requests as per NFR. REST APIs: Easy integration with Weather API & Govt DB.
3	Dat abase / Dat a St orage (e.g., MySQL, MongoDB, Post gr eSQL)	Post gr eSQL for St ruct ured Dat a Redi s for Cachi ng AWS S3 for File St orage	Post gr eSQL: Open- sour ce, handl es 10 lakh+ farmer records wi th st rong rel at i onal mappi ng bet ween Farmer, Soi l, Crop, Market dat a. ACID compl i ant for dat a i nt egr i ty. Redi s: I n- memory cache for weat her API & market prices. Reduces response t i me t o < 2 seconds as per NFR. AWS S3: St ores DFD di agrams, farmer report s, ML model files. 99.9% durabi l i ty & l ow cost .
4	Cl oud / Host i ng / Depl oyment (e.g., AWS, Azure, Heroku, Docker)	AWS EC2 for Comput e Docker + Kuber net es AWS Lambda for Serverl ess	AWS EC2: Aut o- scal i ng t o handl e 50,000 concurrent farmers dur i ng peak sowi ng season. Ensures 99.5% upt i me as per NFR. Docker + K8s: Cl oud- agnost i c depl oyment . Easy mai nt enance & mont hly ML model updat es. AWS Lambda: Cost - effect i ve for SMS gat eway & OTP generat i on. Pay- per- use, i deal for Govt budget .

5	Version Control & CI/CD	Git + GitHub GitHub Actions for CI/CD SonarQube for Code Quality	Git + GitHub: Team collaboration, version history for ML model code & Flutter app. Free for educational use. GitHub Actions: Automated testing + deployment to AWS. Every code push triggers unit tests. Reduces manual errors. SonarQube: Scans code for security vulnerabilities. Ensures DPDP Act 2023 compliance before product release.
	(e.g., GitHub, GitLab, Jenkins)		
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	OpenWeather API Agmarknet API (Govt of India) Twilio SMS Gateway Google Maps API	OpenWeather API: Real-time rainfall, temp, humidity data for crop suggestion accuracy. Free tier supports 1000 calls/day. Agmarknet API: Official Govt source for daily crop market prices from mandis. Ensures profit calculation is authentic. Twilio SMS: Sends crop suggestions to farmers with feature phones. 99.95% delivery rate. Google Maps API: Auto-detects farmer GPS location for region-specific soil data. Supports Telugu language maps.